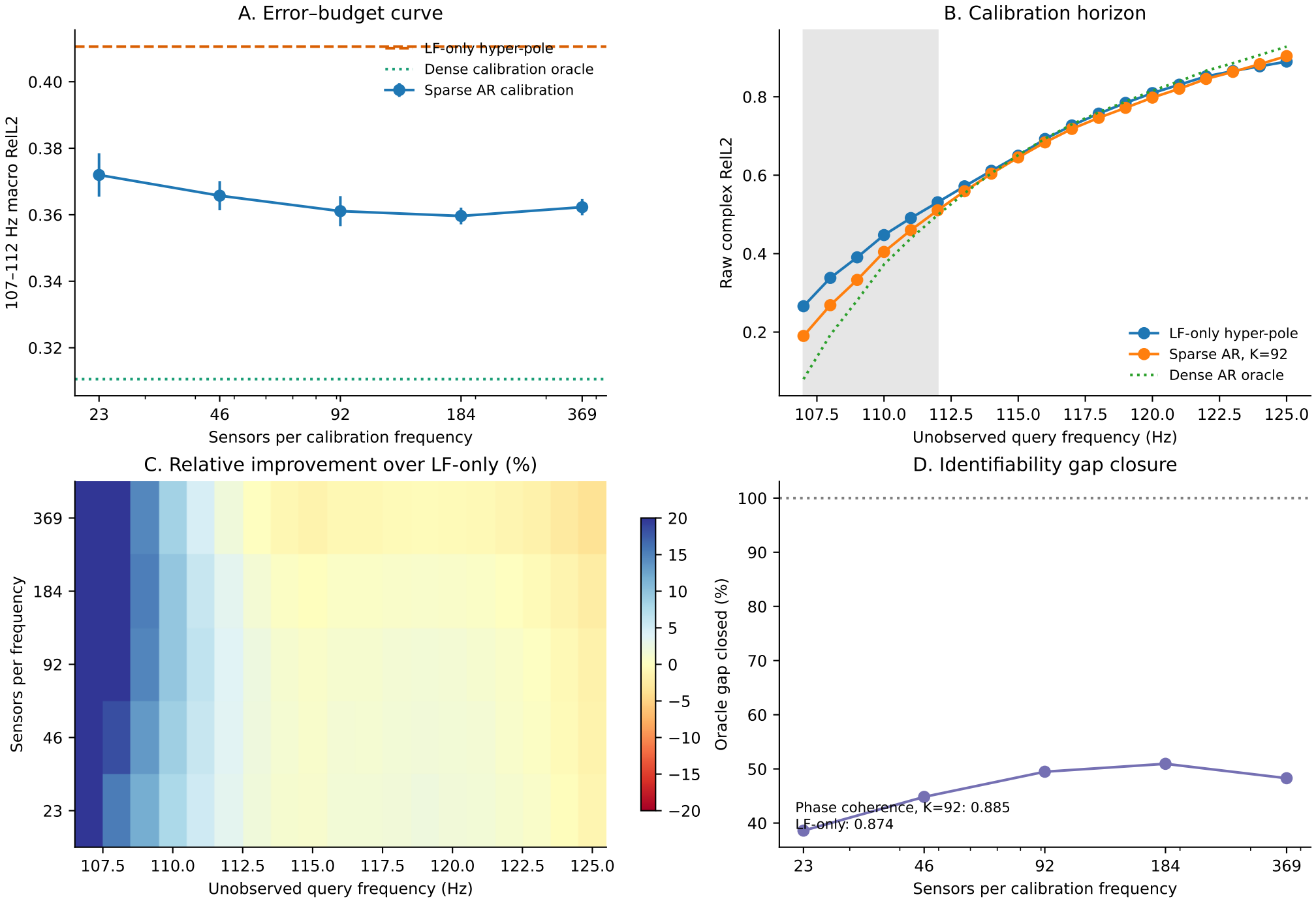
A. Error–budget curve
LF-only hyper-pole
Dense calibration oracle
Sparse AR calibration
107–112 Hz macro RelL2
0.40
0.38
0.36
0.34
0.32
23
46
92
184
369
Sensors per calibration frequency
B. Calibration horizon
Raw complex RelL2
0.8
0.6
0.4
0.2
LF-only hyper-pole
Sparse AR, K=92
Dense AR oracle
107.5
110.0
112.5
115.0
117.5
120.0
122.5
125.0
Unobserved query frequency (Hz)
C. Relative improvement over LF-only (%)
Sensors per frequency
369
184
92
46
23
20
15
10
5
0
−5
−10
−15
−20
107.5
110.0
112.5
115.0
117.5
120.0
122.5
125.0
Unobserved query frequency (Hz)
D. Identifiability gap closure
Oracle gap closed (%)
100
90
80
70
60
50
40
Phase coherence, K=92: 0.885
LF-only: 0.874
23
46
92
184
369
Sensors per calibration frequency